

Krishnaa Vadivel | Embedded Systems Engineer | Firmware | FPGA | Robotics

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Summary

Embedded systems engineer with hands-on experience across MCU firmware, FPGA logic, sensor-logging systems, robotics, and engineering validation. Comfortable moving from low-level C and digital logic through oscilloscope-based debugging, field testing, and host-side tooling for visualization and diagnostics.

Experience

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Developed embedded sensor-logging and control functionality for geological well tools using Lattice FPGA, PIC32, dsPIC, STM32, and C8051-class platforms.
- Supported board- and system-level bring-up with oscilloscope-based debugging, field validation, and targeted PCB updates in Altium.
- Built companion CUDA engineering tools for waveform processing, visualization, and diagnostics around the embedded hardware stack.

Yale University

Embedded Systems Teaching Assistant / PhD Researcher

2021–present

- Supported embedded systems instruction and student lab work involving firmware, hardware interfaces, debugging, and instrumentation.
- Brought a strong embedded perspective into broader scientific-computing and engineering workflows during PhD research.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and delivered APS presentations in 2024, 2025, and 2026, strengthening the research side of my hardware-software profile.

Selected Projects

EMG Robotic Hand: Embedded systems project combining surface EMG sensing, analog filtering, PIC32 control logic, and servo actuation; related work was published in Circuit Cellar.

Maze Mapping Robot: Course-based robotics work involving navigation logic, wireless coordination, and remote-station control.

MIPS Processor: Single-cycle Verilog CPU with self-checking test bench for readable digital-design validation.

Matrix-Multiply Accelerator: Small Verilog accelerator block showing datapath and control concepts relevant to embedded ML hardware.

Education

Yale University

PhD, Electrical Engineering

2021–present

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Technical Skills

Firmware: C, C++, MCU firmware, hardware-software integration

Hardware: Verilog, FPGA, PIC32, dsPIC, STM32, C8051, sensor interfaces

Debug: Oscilloscopes, bench instrumentation, field testing, board bring-up

Software: Python, Linux, CUDA, C#, ASP.NET

Interfaces: SPI, I2C, UART, embedded control and logging workflows